

ECE368: Probabilistic Reasoning

Lab 3: Hidden Markov Model

You can complete this lab in a group of two. Please provide the name and student number of both members.

Name: Bharat Bhargava	Student Number: 1010380892
Name: Daivik Dhar	Student Number: 1010214260

You should hand in: 1) A scanned .pdf version of this sheet with your answers (file size should be under 2 MB); 2) one Python file `inference.py` that contains your code. The files should be uploaded to Quercus.

- (a) Write down the formulas of the forward-backward algorithm to compute the marginal distribution $p(\mathbf{z}_i | (\hat{x}_0, \hat{y}_0), \dots, (\hat{x}_{N-1}, \hat{y}_{N-1}))$ for $i = 0, 1, \dots, N - 1$. Your answer should contain the initializations of the forward and backward messages, the recursion relations of the messages, and the computation of the marginal distribution based on the messages. (0.5 pt)

Let $o_k = (\hat{x}_k, \hat{y}_k)$ denote the observation at time step k .

1. Forward Messages ($\alpha_i(z_i) = p(o_0, \dots, o_i, z_i)$)

Initialization ($i = 0$): $\alpha_0(z_0) = p(z_0)p(o_0|z_0)$

Recursion ($i \geq 1$): $\alpha_i(z_i) = p(o_i|z_i) \sum_{z_{i-1}} \alpha_{i-1}(z_{i-1})p(z_i|z_{i-1})$

2. Backward Messages ($\beta_i(z_i) = p(o_{i+1}, \dots, o_{N-1}|z_i)$)

Initialization ($i = N - 1$): $\beta_{N-1}(z_{N-1}) = 1$

Recursion ($i < N - 1$): $\beta_i(z_i) = \sum_{z_{i+1}} \beta_{i+1}(z_{i+1})p(z_{i+1}|z_i)p(o_{i+1}|z_{i+1})$

3. Marginal Distribution ($\gamma_i(z_i) = p(z_i|o_0, \dots, o_{N-1})$)

The marginal distribution is the normalized product of the forward and backward messages:

$$p(z_i|o_0, \dots, o_{N-1}) = \frac{\alpha_i(z_i)\beta_i(z_i)}{\sum_{z_i} \alpha_i(z_i)\beta_i(z_i)}$$

- (b) After you run the forward-backward algorithm on the data in `test.txt`, write down the obtained marginal distribution of the state at $i = 99$ (the last time step), i.e., $p(\mathbf{z}_{99} | (\hat{x}_0, \hat{y}_0), \dots, (\hat{x}_{99}, \hat{y}_{99}))$. Only include states with non-zero probability in your answer. (1 pt)

State (11, 0, 'stay') with probability 0.8103
State (11, 0, 'right') with probability 0.1796
State (10, 1, 'down') with probability 0.0101

2. Modify your forward-backward algorithm so that it can handle missing observations. After you run the modified forward-backward algorithm on the data in `test_missing.txt`, write down the obtained marginal distribution of the state at $i = 30$, i.e., $p(\mathbf{z}_{30} | (\hat{x}_0, \hat{y}_0), \dots, (\hat{x}_{99}, \hat{y}_{99}))$. Only include states with non-zero probability in your answer. (0.5 pt)

State (6, 7, 'right') with probability 0.9130
 State (5, 7, 'right') with probability 0.0435
 State (5, 7, 'stay') with probability 0.0435

3. (a) Write down the formulas of the Viterbi algorithm using $\mathbf{z}_i$ and $(\hat{x}_i, \hat{y}_i)$, $i = 0, 1, \dots, N - 1$. Your answer should contain the initialization of the messages and the recursion of the messages in the Viterbi algorithm. (0.5 pt)

Let $o_k = (\hat{x}_k, \hat{y}_k)$ denote the observation at time step k . Let $\omega_i(z_i)$ be the maximum probability of a state sequence ending at state z_i , and $\psi_i(z_i)$ be the backpointer.

1. Initialization ($i = 0$):

$$\omega_0(z_0) = p(z_0)p(o_0|z_0)$$

$$\psi_0(z_0) = \text{None}$$

2. Recursion ($i = 1, \dots, N - 1$):

$$\omega_i(z_i) = p(o_i|z_i) \max_{z_{i-1}} [\omega_{i-1}(z_{i-1})p(z_i|z_{i-1})]$$

$$\psi_i(z_i) = \arg \max_{z_{i-1}} [\omega_{i-1}(z_{i-1})p(z_i|z_{i-1})]$$

3. Termination & Backtracking:

$$z_{N-1}^* = \arg \max_{z_{N-1}} \omega_{N-1}(z_{N-1})$$

$$\text{For } i = N - 1, N - 2, \dots, 1: z_{i-1}^* = \psi_i(z_i^*)$$

- (b) After you run the Viterbi algorithm on the data in `test_missing.txt`, write down the last 10 hidden states of the most likely sequence (i.e., $i = 90, 91, 92, \dots, 99$) based on the MAP estimate. (1.5 pt)

$i = 90$: (11, 5, 'down') $i = 95$: (10, 7, 'left')
 $i = 91$: (11, 6, 'down') $i = 96$: (9, 7, 'left')
 $i = 92$: (11, 7, 'down') $i = 97$: (8, 7, 'left')
 $i = 93$: (11, 7, 'stay') $i = 98$: (7, 7, 'left')
 $i = 94$: (11, 7, 'stay') $i = 99$: (6, 7, 'left')

4. Compute and compare the error probabilities of $\{\tilde{\mathbf{z}}_i\}$ and $\{\tilde{\mathbf{z}}_i\}$ using the data in `test_missing.txt`. The error probability of $\{\tilde{\mathbf{z}}_i\}$ is . The error probability of $\{\tilde{\mathbf{z}}_i\}$ is . (0.5 pt)
5. Is sequence $\{\tilde{\mathbf{z}}_i\}$ a valid sequence? If not, please find a small segment $\tilde{\mathbf{z}}_i, \tilde{\mathbf{z}}_{i+1}$ that violates the transition model for some time step i . Your answer should specify the value of i as well as the corresponding states $\tilde{\mathbf{z}}_i, \tilde{\mathbf{z}}_{i+1}$. (0.5 pt)

No, the sequence is not valid.

At $i = 64$, there is an invalid transition from $\tilde{\mathbf{z}}_{64} = (3, 7, \text{'stay'})$ to $\tilde{\mathbf{z}}_{65} = (2, 7, \text{'stay'})$ because the probability of making this specific transition is 0.